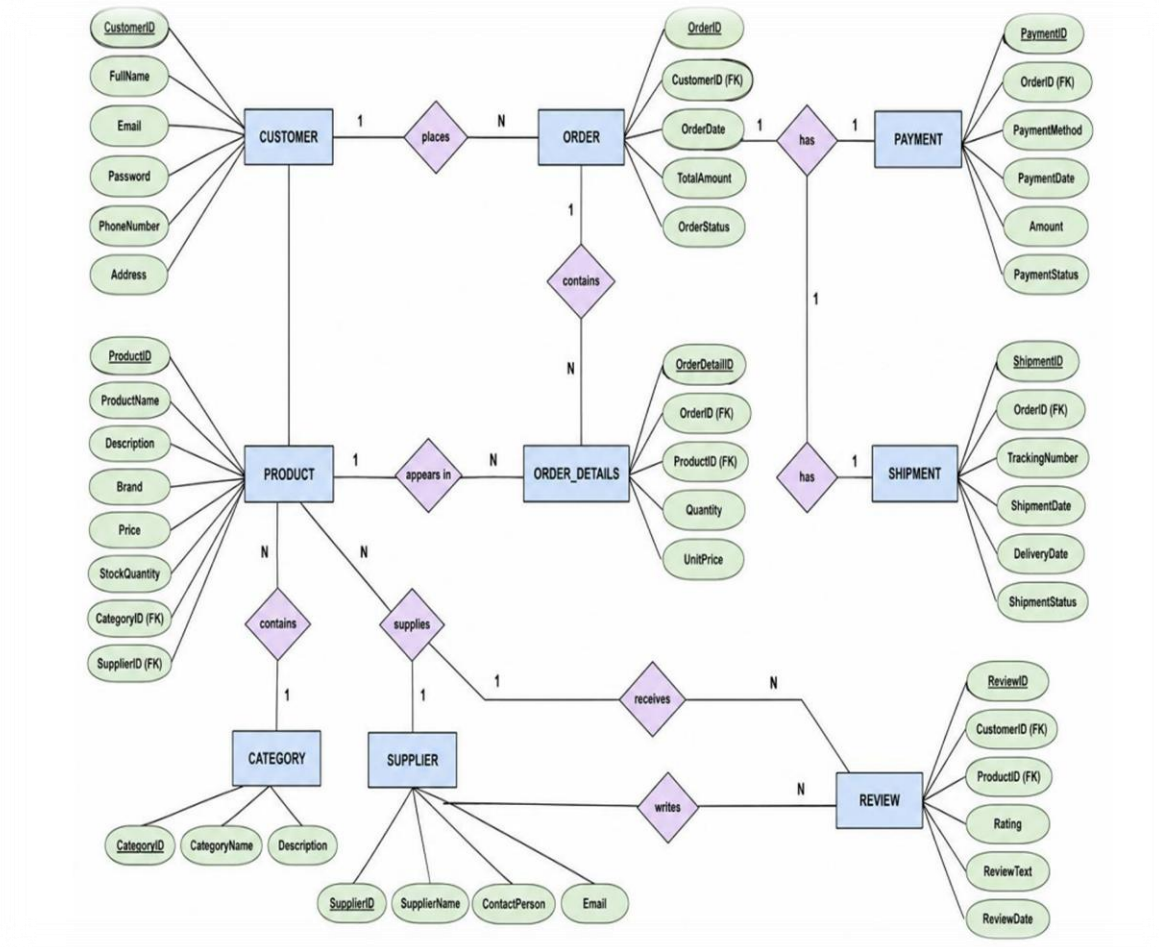


ENTITY RELATIONSHIP (ER) DIAGRAM

E-COMMERCE ORDER MANAGEMENT DATABASE SYSTEM



Explanation of the ER Diagram

1. Entities

- Entities represent real-world objects whose data is stored in the database.
- The entities used in this ER Diagram are:
 - Customer
 - Category
 - Product
 - Supplier
 - Shopping Cart
 - Wishlist
 - Order

- Order_Details
 - Payment
 - Shipment
 - Review
-

2. Attributes

- Attributes are the properties or characteristics of an entity.
- They store information related to each entity.
- Some important attributes used in this ER Diagram are:

- **Customer**

- Customer_ID
 - Customer_Name
 - Email
 - Phone_Number
 - Password
 - Address
-

- **Category**

- Category_ID
 - Category_Name
 - Description
-

- **Product**

- Product_ID
 - Product_Name
 - Brand
 - Price
 - Stock_Quantity
 - Description
 - Category_ID
 - Supplier_ID
-

- **Supplier**

- Supplier_ID
 - Supplier_Name
 - Contact_Number
 - Email
-

- **Shopping Cart**

- Cart_ID
- Customer_ID
- Product_ID

- Quantity
- Total_Price

- **Wishlist**

- Wishlist_ID
- Customer_ID
- Product_ID
- Date_Added

- **Order**

- Order_ID
- Customer_ID
- Order_Date
- Total_Amount
- Order_Status

- **Order_Details**

- OrderDetail_ID
- Order_ID
- Product_ID
- Quantity
- Unit_Price

- **Payment**

- Payment_ID
- Order_ID
- Payment_Method
- Payment_Date
- Amount
- Payment_Status

- **Shipment**

- Shipment_ID
- Order_ID
- Tracking_Number
- Shipment_Status
- Delivery_Date

- **Review**

- Review_ID
- Customer_ID
- Product_ID

- Rating
 - Review_Text
-

3. Relationships

- Relationships describe how different entities are connected with one another.
 - The relationships used in this ER Diagram are:
 - A **Customer** places an **Order**.
 - A **Customer** adds **Products** to the **Shopping Cart**.
 - A **Customer** creates a **Wishlist**.
 - A **Category** contains multiple **Products**.
 - A **Supplier** supplies multiple **Products**.
 - An **Order** contains **Order_Details**.
 - An **Order** has one **Payment**.
 - An **Order** has one **Shipment**.
 - A **Customer** writes a **Review** for a **Product**.
 - A **Product** receives multiple **Reviews**.
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4. Primary Key (PK)

- A **Primary Key** uniquely identifies each record in an entity.
 - Every entity in the database has its own primary key.
Examples:
 - Customer → Customer_ID
 - Product → Product_ID
 - Category → Category_ID
 - Supplier → Supplier_ID
 - Cart → Cart_ID
 - Wishlist → Wishlist_ID
 - Order → Order_ID
 - Order_Details → OrderDetail_ID
 - Payment → Payment_ID
 - Shipment → Shipment_ID
 - Review → Review_ID
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5. Foreign Key (FK)

- A **Foreign Key** is an attribute that creates a relationship between two entities.
- It connects one table with another by referring to the primary key of another entity.
Examples:
 - Customer_ID in **Order**
 - Product_ID in **Order_Details**

- Order_ID in **Payment**
 - Order_ID in **Shipment**
 - Category_ID in **Product**
 - Supplier_ID in **Product**
 - Customer_ID and Product_ID in **Review**
 - Customer_ID and Product_ID in **Shopping Cart**
 - Customer_ID and Product_ID in **Wishlist**
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6. Cardinality

- Cardinality defines how many records of one entity can be associated with another entity.

Examples:

- One Customer → Many Orders (1:N)
 - One Category → Many Products (1:N)
 - One Supplier → Many Products (1:N)
 - One Order → Many Order_Details (1:N)
 - One Order → One Payment (1:1)
 - One Order → One Shipment (1:1)
 - One Product → Many Reviews (1:N)
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